Problem D. Veci

Time Limit	1000 ms
Mem Limit	1048576 kB
OS	Linux

Your program will be given an integer X. Find the smallest number larger than X consisting of the same digits as X.

Input

The first line of input contains the integer X ($1 \leq X \leq 999\,999$). The first digit in X will not be a zero.

Output

Output the result on a single line. If there is no such number, output 0.

Sample 1

Input	Output
156	165

Sample 2

Input	Output
330	0

Sample 3

Input	Output
27711	71127